

AMD Developer Hackathon: ACT II · Unicorn Track



Crop Advisor Lite

AI crop advisory for Pakistan's 8 million smallholder farmers – in English and Urdu.

Meta Llama 4 · Fireworks AI (AMD-hosted inference) · Streamlit · Docker

The problem

- **8M+** smallholder farms in Pakistan (1-12 acres) drive the country's food supply.
- Most farmers have **no affordable access to agronomic advice**: what to sow after harvest, when the sowing window closes, what an acre costs in inputs.
- Government extension services are stretched thin; advice travels by word of mouth – often late, often wrong.
- Bad crop decisions compound: one wrong season can wipe out a smallholder's yearly margin.

The solution – a chat advisor in the farmer's own language

I have 5 acres in Multan, just harvested wheat. Water is short. What should I sow next?

Crop Advisor Lite: "For a 5-acre farm in Multan, sown after wheat harvest, consider: **Cotton** – sowing window May-June... **Maize**... **Mung bean (Moong)**: a drought-tolerant pulse crop. Sowing window: June-July. Total: PKR 5,400-9,800 per acre... cotton requires more water than maize and mung bean. Consult your local Zarai Taraqati office or a qualified agronomist for specific advice."

Actual model output (Meta Llama 4, truncated) – not a mockup.

- Decision-ready answers: named crops, months, rupee ranges – not vague generalities.
- English in, English out. Urdu in, Urdu out.

- Honest about limits: refuses to invent market prices; routes high-stakes calls to extension offices.

Built on AMD

Model	Meta Llama 4 – open-weight, multilingual (handles Urdu natively); any OpenAI-compatible host via <code>API_BASE_URL</code>
Grounding	BM25 retrieval with crop-aware routing over 9 Pakistani agricultural documents (govt policy, PCRWR research, FAO calendar) – answers cite sources inline; Urdu questions are translated internally so they cite too
Evaluation	16/16 on a corpus-grounded eval harness incl. honesty probes (refuses to invent market prices) – reproducible via <code>eval/run_eval.py</code>
Inference	Fireworks AI API – AMD-hosted inference for this hackathon; serverless, pay-per-token
App	Streamlit (Python) – chat UI with example prompts in English and Urdu
Packaging	Docker – one command to run anywhere: <code>docker run -p 8501:8501 -e FIREWORKS_API_KEY=...</code>

What's done – and what's next

- **Done – grounding:** answers cite retrieved passages from 9 Pakistani agricultural documents; no relevant source means the app says so instead of inventing one.
- **Done – measured accuracy:** 16/16 on a reproducible eval harness with honesty and safety probes.
- **Next – Urdu voice:** speech in/out for farmers who can't read or type comfortably.
- **Next – WhatsApp delivery:** meet farmers on the platform they already use daily.
- **Cost path:** open-weight models on AMD-hosted inference keep per-farmer cost near zero – the unit economics of free advice.